

UMAP (Uniform Manifold Approximation and Projection)

UMAP is a **dimensionality reduction and visualization technique**, similar to t-SNE, but usually:

- ✓ Faster
 - ✓ Scales better to large datasets
 - ✓ Preserves both local and global structure better
 - ✓ Can be used for visualization and as a preprocessing step for ML
-

The Party Analogy

Let's compare PCA, t-SNE, and UMAP using the same example.

Imagine 1,000 people at a conference.

People naturally form groups:

- Developers
- Data Scientists
- Managers
- Students

Each person has 50 characteristics.

PCA says

"Let me find the directions with the most variation."

Result:

Developer Student Manager



Clusters may overlap.

t-SNE says

"Keep friends together."

Result:

Developers	Students
●●●●●●	●●●●●●
Managers	Data Scientists
●●●●●●	●●●●●●

Beautiful clusters.

But distances between clusters may not mean much.

UMAP says

"Keep friends together AND preserve the overall neighborhood map."

Result:

Developers	-----	Data Scientists
\		
\		
	Students	----- Managers

You can see both:

- Who belongs together
 - Which groups are closer overall
-

Why Was UMAP Created?

Researchers loved t-SNE because it produced beautiful visualizations.

But t-SNE had issues:

Problem 1

Slow on large datasets.

Example:

10,000 samples

can take significant time.

Problem 2

Global relationships get distorted.

Suppose:

Cats
Dogs
Cars

In reality:

Cats closer to Dogs
Dogs far from Cars

t-SNE may show:

Cats
Dogs
Cars

The spacing can be misleading.

UMAP tries to preserve more of the overall structure.

Intuition Behind UMAP

Imagine a city map.

People are connected by roads.

UMAP first builds a graph:

A ---- B ---- C
 \ |
 \ |
 D----E

Nearby points become connected.

Then UMAP compresses the graph into 2D while trying to preserve those relationships.

Think:

```
Original 100D world
      ↓
Build neighborhood network
      ↓
Compress to 2D
      ↓
Keep structure intact
```

Simple Example

Suppose we have six customers.

Customer Income Spending

A	40K	38K
B	42K	39K
C	41K	40K
D	120K	90K
E	125K	92K
F	123K	89K

Clearly:

Group 1:
A B C

Group 2:
D E F

In high-dimensional space:

```
Income
Spending
Age
Location
Purchases
...
```

perhaps 50 dimensions.

UMAP converts:

```
50D
↓
```

2D

Result:

A B C

D E F

Clusters become visible.

More Realistic Example

Imagine Netflix users.

Each user has:

Movies watched
Genres
Ratings
Viewing time
Language
Devices
...

100+ features.

UMAP may reveal:

Action Fans

●●●●●●

Anime Fans

●●●●●●

Documentary Fans

●●●●●●

Comedy Fans

●●●●●●

Marketing teams can instantly identify audience segments.

Why UMAP Is Popular in AI

Image Embeddings

Suppose a CNN produces:

512 features per image

UMAP reduces:

512

↓

2

Visualization:

Cats

•••••

Dogs

•••••

Cars

•••••

Useful for understanding model behavior.

LLM Embeddings

Suppose you generate embeddings for:

Python

Java

Machine Learning

Cooking

Football

UMAP often produces:

Programming

• • •

Sports

• •

Cooking

•

Semantically similar concepts cluster together.

Important Parameters

1. n_neighbors

Controls neighborhood size.

Small value:

Focus on local structure

Large value:

Focus on global structure

Example:

```
n_neighbors = 15
```

2. min_dist

Controls cluster tightness.

Small:

Tighter clusters

Large:

More spread-out clusters

PCA vs t-SNE vs UMAP

Feature	PCA	t-SNE	UMAP
Fast	✓	✗	✓
Handles non-linear patterns	✗	✓	✓
Preserves variance	✓	✗	Partly
Preserves local structure	Moderate	Excellent	Excellent
Preserves global structure	Good	Poor	Better
Scales to large datasets	Excellent	Poor	Excellent

Feature	PCA	t-SNE	UMAP
ML preprocessing	Common	Rare	Common
Visualization	Good	Excellent	Excellent

Typical Industry Workflow

For a dataset with 10,000 features:

```

10000 Features
  ↓
PCA
  ↓
100 Features
  ↓
UMAP
  ↓
2 Features
  ↓
Visualization

```

This is common in:

- Genomics
 - Customer analytics
 - Fraud detection
 - NLP embeddings
 - Image embeddings
 - LLM vector analysis
-

Example in Single-Cell Biology

Each cell may contain:

```
20,000 gene measurements
```

Researchers apply:

```

20,000 Features
  ↓
PCA
  ↓
50 Components
  ↓
UMAP
  ↓
2D Plot

```

Result:

T Cells

•••••

B Cells

•••••

NK Cells

•••••

Each cluster often represents a different cell type.

This is one of the most famous real-world uses of UMAP.

Interview Answer

UMAP (Uniform Manifold Approximation and Projection) is a non-linear dimensionality reduction technique that constructs a graph representing local relationships in high-dimensional data and then optimizes a lower-dimensional representation while preserving those relationships. It is faster than t-SNE, scales better to large datasets, and generally preserves both local and global data structure more effectively.

Memory Trick

Think of the three techniques this way:

PCA = Preserve Variance

t-SNE = Preserve Friends

UMAP = Preserve Friends + City Map

Or even shorter:

PCA → Compression

t-SNE → Visualization

UMAP → Visualization + Structure

That's why many modern AI, genomics, and embedding-analysis workflows prefer **UMAP** over **t-SNE** today.